DO NOW · AO1 REVISION

Acceleration

Forces
Paper 2 · Combined and Separate
11 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 5 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**State the standard unit of acceleration.**	Metre per second squared (m/s^2)	☐
2 ★	**State what the gradient of a velocity–time graph represents.**	Acceleration	☐
3 ★	**State what the area under a velocity–time graph represents.**	The distance travelled	☐
4	**State the acceleration of an object in free fall near the Earth's surface.**	About 9.8 m/s^2	☐
5 ★	**Define terminal velocity.**	The constant velocity of a falling object when the resistive forces equal its weight.	☐
6	**State the equation linking acceleration, change in velocity and time taken.**	acceleration = change in velocity ÷ time taken ($a = \Delta v \div t$)	☐
7	**State the equation for uniform acceleration.**	$(\text{final velocity})^2 - (\text{initial velocity})^2 = 2 \times \text{acceleration} \times \text{distance}$ ($v^2 - u^2 = 2\,a\,s$)	☐
8	**What is a deceleration?**	A negative acceleration: the object is slowing down.	☐
9	**State what a horizontal line on a velocity–time graph shows.**	Constant velocity.	☐
10	**State what a straight, sloping line on a velocity–time graph shows.**	Constant acceleration.	☐
11 ★	**What is acceleration?**	The rate of change of velocity.	☐

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